FRANCEGO VIP TRANSFERS & TOURS

VPS Server Security & Production Deployment Guide

Complete systems manual for deploying on dedicated Linux VPS (Ubuntu/Debian) with PostgreSQL, Nginx, Let's Encrypt SSL, and server-end security hardening.

LINUX SYSADMIN MANUAL | CIS HARDENING COMPLIANT | POSTGRESQL NATIVE

Server Hostname: srv2088392
Target Domain: francegotransfers.io
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1. Server-End Security & Sudoer Hardening

Granting `sudo` permissions to a dedicated non-root user (e.g. `francegotransfers`) is standard industry best practice. However, running applications as root or leaving server ports exposed creates significant security hazards. Follow this checklist to ensure complete server security.

Security Architecture Principles

Security Domain	Standard Best Practice	Vulnerable Pattern to Avoid
Application Runtime	Run PM2 and Node.js strictly under `francegotransfers`	Running `pm2 start` as `root`
Sudo Configuration	Enforce password prompt for all `sudo` actions	Using `NOPASSWD: ALL` in `/etc/sudoers`
Secret Storage	Lock down `.env` file permissions to `600`	Leaving `.env` world-readable (`644` or `777`)
Database Access	Bind PostgreSQL solely to `127.0.0.1`	Exposing port `5432` to `0.0.0.0/0`
Firewall (UFW)	Only open Ports 22 (SSH), 80 (HTTP), 443 (HTTPS)	Exposing Next.js port `3000` publicly
SSH Authentication	Disable root SSH login (`PermitRootLogin no`)	Permitting direct SSH logins as root

Server Hardening Commands

1. Restrict Secret File Permissions

```
chmod 600 /var/www/francego/.env
```

2. Configure Linux Directory & File Permissions

```
# Ensure correct user ownership
sudo chown -R francegotransfers:francegotransfers /var/www/francego

# Directory permissions: 755 (drwxr-xr-x)
sudo find /var/www/francego -type d -exec chmod 755 {} \;

# File permissions: 644 (-rw-r--r--)
sudo find /var/www/francego -type f -exec chmod 644 {} \;

# Re-lock .env
sudo chmod 600 /var/www/francego/.env
```

3. Configure UFW Firewall

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp    # SSH
sudo ufw allow 80/tcp    # HTTP (Nginx)
sudo ufw allow 443/tcp   # HTTPS (Nginx SSL)
sudo ufw enable
```

4. Disable Direct Root SSH Login

Edit `/etc/ssh/sshd_config` and verify:

```
PermitRootLogin no
MaxAuthTries 4
```

```
sudo systemctl restart ssh
```

5. Install Fail2ban (Anti-Brute Force)

```
sudo apt install -y fail2ban
sudo systemctl enable fail2ban && sudo systemctl start fail2ban
```

2. Step-by-Step VPS Production Deployment

Step 1: Domain DNS A Record Setup

Point your domain's DNS A records to your VPS IP address (`141.136.36.40`):

Type	Host	Value (VPS IP)	TTL
`A`	`@`	`141.136.36.40`	Automatic / 300s
`A`	`www`	`141.136.36.40`	Automatic / 300s

Step 2: PostgreSQL Installation & Database Creation

```
sudo apt update && sudo apt install -y postgresql postgresql-contrib
sudo systemctl enable postgresql && sudo systemctl start postgresql

# Create Database User and Schema
sudo -u postgres psql
```

Inside PostgreSQL:

```
CREATE USER francego_user WITH PASSWORD 'YourSecurePassword2026!';
CREATE DATABASE francego_db OWNER francego_user;
GRANT ALL PRIVILEGES ON DATABASE francego_db TO francego_user;
\q
```

Step 3: Import `database_init.sql`

Import the master SQL script containing all 6 tables, constraints, 5 categories, 14 hubs, and 25 routes:

```
cd /var/www/francego
psql -U francego_user -d francego_db -f database_init.sql
```

Step 4: Configure Production `.env`

```
DATABASE_URL="postgresql://francego_user:YourSecurePassword2026!@127.0.0.1:5432/francego_db?schema=public"
ADMIN_EMAIL="brian.arangala@outlook.com"
ADMIN_PASSWORD="YourSecureAdminPassword2026!"
JWT_SECRET="8e30d232bfe7d16e9e4c1c3206268063d9cfdb2ea2ff8a31"
PARTNER_API_KEY="your_b2b_secret_partner_key_2026"
SMTP_HOST="smtp-mail.outlook.com"
SMTP_PORT="587"
SMTP_SECURE="false"
SMTP_USER="projet15df@outlook.fr"
SMTP_PASS="your-email-app-password"
ADMIN_DISPATCH_EMAIL="projet15df@outlook.fr"
```

Step 5: Build & Run Next.js Application

```
cd /var/www/francego
git pull origin production-postgres-manual
npx prisma generate
npm run build

# Start via PM2
pm2 start npm --name "francego" -- start -- -p 3000
pm2 save && pm2 startup
```

Step 6: Nginx Reverse Proxy & Free Let's Encrypt SSL

```
sudo nano /etc/nginx/sites-available/francego
```

```
server {
    listen 80;
    server_name francegotransfers.io www.francegotransfers.io;
    client_max_body_size 20M;

    location / {
        proxy_pass http://127.0.0.1:3000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

```
sudo ln -s /etc/nginx/sites-available/francego /etc/nginx/sites-enabled/
sudo nginx -t && sudo systemctl reload nginx

# Install SSL
sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d francegotransfers.io -d www.francegotransfers.io
```